

COMPUTER NETWORKS

Assessment - 1

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Course } : CSA0702
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Slot : B

Test : Short Answers

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Total marks : 20

obtained marks : 18

File Transfer

- File Size = 150 MB = 150,000,000 bytes
- Segment Size = 1500 bytes

Number of segments (Frames)

$$= 150,000,000 \div 1500$$

$$= 100,000 \text{ segments}$$

Data link overhead

$$= 100,000 \times 40 = 4,000,000 \text{ bytes (4 MB)}$$

Total data transmitted

$$= 150 \text{ MB} + 4 \text{ MB} = 154 \text{ MB}$$

Transmission Time

$$154 \text{ MB} = 1232 \text{ Mb}$$

$$\text{Time} = 1232 \div 50 = 24.64 \text{ seconds}$$

Explanation:

→ Transport layer divides data, performs flow / error control.

→ Data link layer adds headers / trailers, detects errors, and delivers frames reliably.

2. sliding window

→ window size = 6

→ total frames = 48

Windows required

$$= 48 \div 6 = 8 \text{ windows}$$

Retransmissions

= 1 lost frame per window

= 8 retransmission

Explanation:

Flow control prevents the sender from overwhelming the receiver, improving efficiency and reducing congestion.

3. IP fragmentation

→ Packet size = 12,000 bytes

→ MTU = 1500 bytes

→ IP header = 20 bytes

Payload per fragment = $1500 - 20 = 1480$ bytes

Fragments required

$$= \text{ceiling}(12000 \div 1480)$$

= 9 fragments

Header overhead

$$= 9 \times 20 = 180 \text{ bytes}$$

Explanation:

The network layer fragments packets and reassembles them at the destination using fragment offset and identification fields.

4.

Sliding window

→ window size = 6

→ Total frames = 48

Windows required

$$= 48 \div 6 = 8 \text{ windows}$$

Retransmissions

= 1 lost frame per window

= 8 retransmissions

Explanation:

Flow control improves communication by matching transmission speed with receiver capacity.

5. Session layer

File = 600 MB

checkpoint every 60 MB

Failure after 390 MB

checkpoints before failure

= 60, 120, 180, 240, 300, 360

= 6 checkpoints

Last checkpoint = 360 MB

Data to retransmit

= $390 - 360$

= 30 MB

Explanation :

Synchronization checkpoint allow transmission to resume from the last checkpoint instead of restarting.

6. Presentation layer

original file = 900 MB

compressed size

= $900 \times 60\%$

= 540 MB

Encrypted Size

$$= 540 \times 1.05$$

$$= 567 \text{ MB}$$

Reduction

$$= (1900 - 567) \div 1900 \times 100$$

$$= 37.1$$

Explanation :

compression reduces transmission size, while encryption ensures data security.

7. FTP Transfer

$$\text{Data} = 3 \text{ GB} = 3000 \text{ MB}$$

$$\text{Request Size} = 3 \text{ MB}$$

Requests

$$= 3000 \div 3$$

$$= 1000 \text{ requests}$$

$$\text{Data} = 24,000 \text{ MB}$$

Transmission time

$$= 24000 \div 120$$

$$= 200 \text{ seconds}$$

Explanation :

The Application layer provide user service such as file transfer, authentication & error.

8. End-to-End Delay

processing delay :

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data link = 2 ms

Physical = 1 ms

Total Processing = 15 ms

Propagation delay = 18 ms

Transmission delay = 12 ms

Total delay

= 15 + 18 + 12

= 45 ms

Explanation :

Each OSI layer processes data before it reaches the Physical layer for transmissions.

Frame overhead

useful data = 80MB = 80,000,000 bytes

Frame Size = 1600 bytes.

overhead = 80 bytes

useful data / frame = $1600 - 80 = 1520$ bytes

Frames required

= ceiling $(80,000,000 \div 1520)$

= 52,632 frames

Total overhead

= $52,632 \times 80$

= 4,210,560 bytes ≈ 4.21 MB

Transmission efficiency

= $(1520 \div 1600) \times 100$

= 95%.

Explanation:

protocol overhead consumes bandwidth, slightly reducing transmission efficiency.

10.

Hospital Network

original file = 240 MB

compression = 30%.

compressed size

$$= 240 \times 70\%$$

$$= 168 \text{ MB}$$

compressed data

$$= 168,000,000 \text{ bytes}$$

segment size = 1400 bytes

segments

$$= 168,000,000 \div 1400$$

$$= 120,000 \text{ segments}$$

overhead / frame = 60 bytes

Protocol overhead

$$= 120,000 \times 60$$

$$= 7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

Final data transmitted

$$= 168 \text{ MB} + 7.2 \text{ MB}$$

$$= 175.2 \text{ MB}$$

Explanation:

→ Presentation layer compresses data.

→ Transport layer segments data

→ Data link layer adds frame overhead